Project Title: **Self-Diagnosis on Social Media: Trends, Risks, and Behavioral Analysis**

Problem Statement: In recent years, social media platforms such as TikTok and Twitter have seen a rise in mental health-related content. Users frequently self-diagnose conditions such as ADHD, anxiety, and depression based on short-form content. This project aims to analyze the trend, engagement, and potential risks of self-diagnosis behavior across social media platforms. The focus will be on data analysis, exploring user behavior, sentiment, and content patterns to generate actionable insights.

Week 1 – Data Collection & Preprocessing Objective: Build a clean and structured dataset for analysis.

1.Collect social media posts: - Twitter posts using hashtags related to ADHD, anxiety, depression. - TikTok videos with relevant hashtags. - Google Trends data for keyword popularity over time.

2. Create a unified data model combining: - Text content - Engagement metrics (likes, shares, comments) - Hashtags - Timestamps and platform info

3. Clean and preprocess text data: - Remove stopwords, punctuation, special characters - Tokenization and lemmatization - Handle missing values and duplicates

Tools: Python (pandas, NLTK, spaCy), SQL Deliverables: Cleaned dataset + preprocessing notebook

Week 2 – Exploratory Analysis (Focus on Trends & Insights) Objective: Understand user behavior and content patterns.

1.Keyword Analysis: Identify most common words and hashtags.

2. Sentiment Analysis: Detect overall sentiment (positive, negative, neutral) in posts.

3. Temporal Analysis: Track trends over time (daily, weekly, monthly).

4. Content Patterns: - Word clouds to visualize common topics - Hashtag clustering to find related topics or communities

5. Generate descriptive statistics: - Average engagement per sentiment/hashtag - Top influencers or users posting about these topics

Tools: Python (pandas, matplotlib, seaborn), NLP libraries Deliverables: Charts and graphs showing trends and insights + summary of key observations

Week 3 – Deep Dive Analysis & Forecasting Objective: Identify drivers behind self-diagnosis and predict future trends.

1. Classify posts as: - Scientific/verified vs misleading/inaccurate

2. User segmentation: - Influencers, experts, general public

3. Correlation analysis: - How engagement relates to sentiment or misinformation
4. Trend forecasting: - Predict keyword popularity and post volume using ML models (e.g., time series forecasting)

5. Generate heatmaps, topic clusters, and insights on potential risks

Tools: Python (scikit-learn, statsmodels) Deliverables: Heatmaps, topic clusters + forecasting plots and analysis report

Week 4 – Dashboard & Final Presentation Objective: Present findings in an interactive, user-friendly way.

1. Build an interactive Tableau dashboard: - Filters: platform, sentiment, keyword, date range - Visualizations: trend lines, engagement metrics, sentiment distribution
2. Prepare a final report summarizing all analyses and findings

3. Create a presentation highlighting: - Key trends and insights - Risks of self-diagnosis - Recommendations for users and platform moderators

Tools: Tableau, MS PowerPoint Deliverables: Interactive dashboard + final report + presentation slides